

Pratyush Khengle

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PROFESSIONAL SUMMARY

Agentic AI Engineer shipping production **multi-agent systems**. Built **Sherlock** at Chubb (4-agent LangGraph auto-debugging **500+ Databricks jobs**) and **VibeApply** (autonomous end-to-end job application agent) within my first year out of **IIT Madras**.

EDUCATION

Indian Institute of Technology Madras

2021 – 2025

B.Tech | CGPA: 8.14

Chennai, India

EXPERIENCE

Machine Learning & Operations Engineer

Jun 2025 – Present

Chubb

Bangalore, India

Sherlock — Autonomous MLOps Debug Platform

- **Self-healing system** that takes a failing Databricks job to a merge-ready PR with **zero human intervention** — onboarded across **500+ Databricks jobs** at Chubb, cutting debug-to-fix from **3–4 hours of triage to under 5 minutes**
- Designed a **4-agent LangGraph pipeline** — stateful debug agent (**6 custom tools**), evaluator scoring **evidence / alignment / fix-specificity / consistency** and routing the next action, Databricks-sandbox verifier, and PR agent — separating diagnosis, judgment, validation, and application so each step can fail independently
- Engineered a sandbox step that executes proposed fixes on a **live Databricks cluster**, generating failure-reproducing tests from the debug agent's RCA, turning autonomous code generation from a leap of faith into a measured outcome
- Gated autonomy on a **confidence-routed policy** — evaluator scores $\geq 75\%$ trigger an autonomous PR; lower scores downgrade to a team-routed suggestion (dev / DE / wheels-source / ops) — self-selecting between action and advice
- Hardened for 24/7 operation with **LangGraph stateful checkpointing** on Azure SQL (resumes mid-investigation after crashes) and **agentic RAG over ChromaDB** retrieving past resolutions for the same error and repo as few-shot context

AI Underwriting Platform

- Developed a **FastAPI web intelligence service** for underwriting analysis, reducing manual research time by **40%**
- Built **web data extraction pipelines using Zyte and SERP APIs** for contextual business intelligence
- Implemented **LLM-based agent workflows** combining web retrieval and structured questionnaire answering
- Integrated services through **Kafka-based asynchronous pipelines**, reducing end-to-end latency by **35%**

AI / LLM Research Intern

Dec 2023 – Jan 2024

Reomnify

Remote

- Engineered a **production RAG pipeline** indexing **200+ documents** for structured **LLM-based QA**, cutting manual cross-document research time by **70%** for the research team
- Expanded document corpus by **3×** via **OCR-based PDF ingestion** using Nanonets, unlocking extraction from image-only documents that text-based search had previously left out of the retrieval index
- Delivered **sub-2 second** end-to-end semantic retrieval using **Sentence Transformers + ChromaDB**, supporting structured QA across **5+ document categories** via both HuggingFace and OpenAI models

PROJECTS

VibeApply — Two-Phase Agentic Job Discovery & Application Pipeline

2026

- *If you're reading this resume, there's a good chance VibeApply is what put it in front of you and likely typed the form, too.*
- **Autonomous job-application agent** that scrapes ranked openings against a resume, then drives a **CDP-attached browser** to fill applications across **Workday, LinkedIn, Wellfound, Lever, Greenhouse, Indeed** — with a **human-in-the-loop review mode** that fills every field and stops at Submit, and an autonomous mode for end-to-end submission with **stale-state recovery** between batches, anti-bot pacing across runs, and per-job audit logs
- Designed a **LangGraph discovery agent** (Claude Haiku 4.5) with **3 custom tools** (multi-board scrape, Tavily startup-board search, structured submit gate) running a **per-specialty 18×4 search matrix** across 10 cities + 6 regional remote markets, with **canonical-URL dedup** against an applied-history index so the same posting never resurfaces, plus a scratch-file pattern keeping descriptions off-context within Haiku's 200K context window
- Engineered a **browser-use apply agent** with custom CDP tools that **pierce shadow DOM** for file upload on Lit / web-component portals, **fold portal redirect tabs** back into the original session, and **set values via native React/Vue setters** — recovering portals where the default browser-use primitives silently fail

TECHNICAL SKILLS

Programming: Python, C++, JavaScript

AI Systems: Agentic AI, Multi-Agent Orchestration, LLMs, RAG, Tool-Calling, Prompt Engineering

LLM Tools: LangGraph, OpenAI API, Gemini API, Model Context Protocol (MCP)

Backend & Systems: FastAPI, Node.js, Kafka, REST APIs, Asynchronous Pipelines

Frontend: React.js, Next.js, Tailwind CSS

Data & Retrieval: Vector Databases, Embeddings, ChromaDB

Databases: SQL, MongoDB, Supabase